CHANDAN SINGH

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PROFESSIONAL SUMMARY

B.Tech CSE student (Expected 2027, CGPA 8.3/10 through 6th semester) specializing in AI/ML engineering with hands-on experience building production-grade computer vision, NLP, and full-stack AI systems. Delivered 6 deployable projects spanning YOLOv8, LSTM, XGBoost, SegFormer, Random Forest, and LLM APIs; competed in national and AMD-sponsored hackathons targeting $20,000+ prize pools.

EDUCATION

B.Tech, Computer Science & Engineering — Expected 2027
Maharishi University of Information Technology, Uttar Pradesh, India | CGPA: 8.3 / 10 (through 6th semester)

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL
AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, segmentation-models-pytorch, SHAP
Backend & Cloud: FastAPI, Flask, Docker, Git, HuggingFace Hub, ROCm, Alembic
Full-Stack & DB: React, Vite, PostgreSQL, SQLAlchemy, JWT Authentication, Twilio API
Concepts: Computer Vision, NLP, Semantic Segmentation, Predictive ML

TECHNICAL PROJECTS

NikaAI – Industrial Defect Detection & Quality Intelligence Platform | *github.com/chandanNegi39671/NikaAI*

- Engineered a production-grade computer vision platform for the AMD AI Hackathon; fine-tuned YOLOv8s achieving 83% mAP across 17 defect classes on 29,354 images and deployed via FastAPI, Celery, and Vercel.
- Implemented Monte Carlo Dropout uncertainty quantification (T=30 passes) to flag low-confidence predictions and a Gemma LLM reasoning layer for automated root-cause analysis, secured with JWT-based RBAC.

HR-RAG-Assistant – Retrieval-Augmented HR Policy Q&A | *github.com/chandanNegi39671/HR-RAG-Assistant*

- Built a full RAG pipeline (LangChain, FAISS, Jina embeddings, Groq LLM) answering natural-language queries over HR policy documents via a LangChain agent tool with a Streamlit chat UI.
- Architected a modular pipeline (loader, chunker, embeddings, vector store, agent) with a persisted FAISS index for instant reloads and tunable chunk size / top-k retrieval.

TRINETRA – AI Scam Intelligence Engine | *github.com/chandanNegi39671/TRINETRA*

- Developed a real-time fraud detection platform achieving 95%+ accuracy using an XGBoost ensemble model to analyze URLs, SMS, and WhatsApp screenshots under low latency.
- Integrated Google Safe Browsing, IP geolocation, and domain entropy scoring into a 0–100 threat score; shipped a full-stack app with a Twilio WhatsApp bot, rate limiting, and input sanitization.

MedVoice-AI – Agentic Clinical Documentation Backend | *github.com/chandanNegi39671/MedVoice-AI*

- Developed a FastAPI backend for outpatient clinical documentation — visit records, audio transcription, EMR generation, consent logging, and PDF export — using hosted OpenAI/Sarvam transcription and HuggingFace EMR generation.
- Designed a bounded agentic decision layer on Groq's API (transcription-quality gate, EMR self-check, triage agent) with fail-safe defaults for judgment-based checks.

Other Projects

- Optimized Falcon Offroad Semantic Segmentation — DeepLabV3+ / mit_b2, achieving 72% mIoU on Duality AI's desert digital-twin dataset (HackWithMumbai 2.0).
- Delivered SIGNSPEAK — word-level ASL recognition using stacked LSTM + MediaPipe, with React/Flask real-time captioning.
- Automated HABS — a healthcare appointment booking system (FastAPI + React) with Random Forest no-show prediction and SHAP explainability.

ACHIEVEMENTS & HACKATHONS

- Deployed NikaAI, a production computer vision platform, as an AMD AI Hackathon participant targeting a $20,000+ prize pool.
- Achieved a top segmentation result at HackWithMumbai 2.0 by building the Falcon offroad semantic segmentation model on Duality AI simulation data.
- Shipped production-grade AI systems using Docker, ROCm, GPU acceleration, and HuggingFace Hub.
- Achieved 95%+ accuracy in a real-time fraud detection system using ensemble ML models.